Design and Verification of AXI4-Lite Slave Interface Protocol using System Verilog

(by CH. ANIRUDH)

1. Introduction

The AXI4-Lite (Advanced Extensible Interface - Lite) is a subset of the AMBA AXI4 protocol designed for simple, low-bandwidth memory-mapped communication in modern SoC (System-on-Chip) architectures. It provides a lightweight, single transaction, register-based communication interface, making it ideal for scenarios where complex burst transactions are not required. AXI4-Lite is commonly used in peripheral control registers, embedded systems, and low-speed data transfers within larger AXI-based interconnects.

Unlike the full AXI4 protocol, AXI4-Lite supports only a single beat per transaction, meaning no burst operations are allowed. This simplicity reduces the complexity of slave peripherals, making it a preferred choice for memory-mapped slave devices in FPGA and ASIC designs. The protocol also ensures independent read and write channels, allowing simultaneous transactions while maintaining efficient control over the bus.

This project focuses on the design and implementation of an AXI4-Lite Slave module using Verilog/System Verilog. The implementation includes RTL coding, verification, simulation, and synthesis to ensure functional correctness and optimized resource utilization. A structured testbench is also developed to verify the protocol compliance, making use of System Verilog verification techniques such as interfaces, transactions, and synchronization mechanisms.

2. AXI4-Lite Protocol Overview

The AXI4-Lite protocol is a simplified version of AXI4, providing a straightforward interface for register-mapped accesses. It is mainly used for accessing control registers and simple peripherals where high-performance burst transactions are not needed. The protocol divides communication into independent read and write channels, ensuring data integrity and pipeline efficiency.

Key Features of AXI4-Lite:

- Simple address-based read and write transactions
- No burst support (single data transaction per request)
- Separate read and write address/data channels
- Uses a handshaking mechanism (VALID & READY signals)
- Ideal for low-power and low-bandwidth applications

Each transaction follows a request-response mechanism, where the master initiates the request, and the slave responds accordingly. The interface consists of five main channels:

1. Address Channel – Transfers the write address from the Write master to the slave.
2. Write Data Channel – Transfers the data to be written.
3. Write Response Channel – Sends an acknowledgment for the write transaction.
4. Read Address Channel – Transfers the read address to the slave.
5. Read Data Channel – Sends the requested data to the master.

Unlike AXI4, where burst transactions are supported, AXI4-Lite ensures only a single address and data transfer per transaction, reducing the complexity of the protocol. This makes it ideal for memory-mapped control registers and peripheral interfaces such as GPIOs, UART controllers, and timers.

Another critical feature of AXI4-Lite is its valid-ready handshake mechanism, which ensures proper synchronization between the master and the slave. Each transaction occurs only when both VALID and READY signals are asserted, ensuring data integrity. This feature helps in designing robust, low-latency communication between master and slave components.

This protocol is widely used in FPGA-based SoCs, embedded processors, and custom ASIC implementations due to its simplicity and efficiency. It enables designers to implement low-overhead interconnects while maintaining compatibility with AXI-based interconnect architectures.

3. AXI4-Lite Design Implementation

The design of the AXI4-Lite Slave module follows a modular approach, ensuring proper functionality and ease of verification. The implementation involves developing register-based memory mapping, where an AXI4-Lite Master can read from and write to registers within the slave module.

Design Approach:

- The AXI4-Lite Slave receives read and write requests from the master and responds accordingly.
- It includes address decoding logic, ensuring correct data transfer to the requested register location.

- The module ensures synchronization between control signals (VALID, READY, RVALID, WVALID).
- Read/write operations are implemented using combinational and sequential logic to ensure accurate timing and data integrity.

The RTL implementation consists of:

- Write Address Decoder – Captures the write address and control signals.
- Write Data Handler – Stores the received data into internal registers.
- Read Address Handler – Fetches the requested data from registers.
- Read Data Response – Sends the appropriate response to the master.

In this design, FSM (Finite State Machine) logic is not used, as the implementation follows a combinational and sequential logic-based approach to manage data transfers efficiently. The design is implemented in Verilog/System Verilog, ensuring modularity and reusability.

RTL CODE:

```
 module axi4_lite_slave #(
    parameter ADDR_WIDTH = 32,
    parameter DATA_WIDTH = 32
)(
    input logic ACLK,
    input logic ARESETn,

    // Write Address Channel
    input logic [ADDR_WIDTH-1:0] AWADDR,
    input logic AWVALID,
    output logic AWREADY,

    // Write Data Channel
    input logic [DATA_WIDTH-1:0] WDATA,
    input logic [DATA_WIDTH/8-1:0] WSTRB,
    input logic WVALID,
        output logic WREADY,

    // Write Response Channel
    output logic [1:0] BRESP,
    output logic BVALID,
    input logic BREADY,

    // Read Address Channel
    input logic [ADDR_WIDTH-1:0] ARADDR,
    input logic ARVALID,
      output logic ARREADY,

    // Read Data Channel
```

```
    output logic [DATA_WIDTH-1:0] RDATA,
    output logic [1:0] RRESP,
    output logic RVALID,
    input logic RREADY
);

    // Internal registers
    logic [DATA_WIDTH-1:0] mem [0:255]; // 256 x 32-bit memory
    logic [1:0] resp;

    initial begin
    for (int i = 0; i < 256; i++) begin
        mem[i] = 32'h0000_0000;
    end
end

    // Write Address Channel
    always_ff @(posedge ACLK or negedge ARESETn) begin
        if (!ARESETn) begin
            AWREADY <= 1'b0;
        end else if (AWVALID && !AWREADY) begin
            AWREADY <= 1'b1;
        end else begin
            AWREADY <= 1'b0;
        end
    end

    // Write Data Channel
    always_ff @(posedge ACLK or negedge ARESETn) begin
        if (!ARESETn) begin
```

```
            WREADY <= 1'b0;
        end else if (WVALID && !WREADY) begin
            WREADY <= 1'b1;
            // Write data to memory
            for (int i = 0; i < DATA_WIDTH/8; i++) begin
                if (WSTRB[i]) begin
                    mem[AWADDR][i*8 +: 8] <= WDATA[i*8 +: 8];
                end
            end
             $display("Memory Write: Addr = %h, Data = %h", AWADDR,
WDATA);
        end else begin
            WREADY <= 1'b0;
        end
    end

    // Write Response Channel
    always_ff @(posedge ACLK or negedge ARESETn) begin
        if (!ARESETn) begin
            BVALID <= 1'b0;
            BRESP <= 2'b00; // OKAY response
        end else if (WREADY && WVALID) begin
            BVALID <= 1'b1;
        end else if (BREADY && BVALID) begin
            BVALID <= 1'b0;
        end
    end

    // Read Address Channel
    always_ff @(posedge ACLK or negedge ARESETn) begin
```

```
        if (!ARESETn) begin
            ARREADY <= 1'b0;
        end else if (ARVALID && !ARREADY) begin
            ARREADY <= 1'b1;
        end else begin
            ARREADY <= 1'b0;
        end
    end

    // Read Data Channel
    always_ff @(posedge ACLK or negedge ARESETn) begin
        if (!ARESETn) begin
            RVALID <= 1'b0;
            RDATA <= '0;
            RRESP <= 2'b00; // OKAY response
        end else if (ARREADY && ARVALID) begin
            RVALID <= 1'b1;
            RDATA <= mem[ARADDR];
            $display("Memory Read: Addr = %h, Data = %h", ARADDR,
mem[ARADDR]);
        end else if (RREADY && RVALID) begin
            RVALID <= 1'b0;
         end
    end

endmodule
```

Testbench:

```
interface axi4_lite_if (input logic ACLK, input logic ARESETn);
    logic [31:0] AWADDR;
    logic AWVALID;
    logic AWREADY;

    logic [31:0] WDATA;
    logic [3:0] WSTRB;
    logic WVALID;
    logic WREADY;

    logic [1:0] BRESP;
    logic BVALID;
    logic BREADY;

    logic [31:0] ARADDR;
    logic ARVALID;
    logic ARREADY;


       logic [31:0] RDATA;
    logic [1:0] RRESP;
    logic RVALID;
    logic RREADY;
endinterface


class transaction;
    rand logic [31:0] addr;
    rand logic [31:0] data;
```

```
    rand logic [3:0] strb;
    rand logic is_write;
endclass

class driver;
    mailbox drv_mbx;
    virtual axi4_lite_if vif;
task run();
    forever begin
        transaction tr;
        drv_mbx.get(tr);
        if (tr.is_write) begin
            $display("Driver: Write Transaction. Addr = %h, Data = %h",
tr.addr, tr.data);
            vif.AWADDR <= tr.addr;
            vif.AWVALID <= 1'b1;
            vif.WDATA <= tr.data;
            vif.WSTRB <= tr.strb;
            vif.WVALID <= 1'b1;
            @(posedge vif.ACLK);
            while (!vif.AWREADY || !vif.WREADY) @(posedge vif.ACLK);
            vif.AWVALID <= 1'b0;
            vif.WVALID <= 1'b0;
            @(posedge vif.ACLK);
            while (!vif.BVALID) @(posedge vif.ACLK);
            vif.BREADY <= 1'b1;
            @(posedge vif.ACLK);
            vif.BREADY <= 1'b0;
              end else begin
            $display("Driver: Read Transaction. Addr = %h", tr.addr);
```

```
            vif.ARADDR <= tr.addr;
            vif.ARVALID <= 1'b1;
            @(posedge vif.ACLK);
            while (!vif.ARREADY) @(posedge vif.ACLK);
            vif.ARVALID <= 1'b0;
            @(posedge vif.ACLK);
            while (!vif.RVALID) @(posedge vif.ACLK);
            vif.RREADY <= 1'b1;
            @(posedge vif.ACLK);
            vif.RREADY <= 1'b0;
        end
    end
endtask

endclass


class monitor;
    mailbox mon_mbx;
    virtual axi4_lite_if vif;

    task run();
        forever begin
            transaction tr;
            @(posedge vif.ACLK);
                    if (vif.AWVALID && vif.AWREADY) begin
                tr = new();
                tr.addr = vif.AWADDR;
                tr.is_write = 1'b1;
                mon_mbx.put(tr);
```

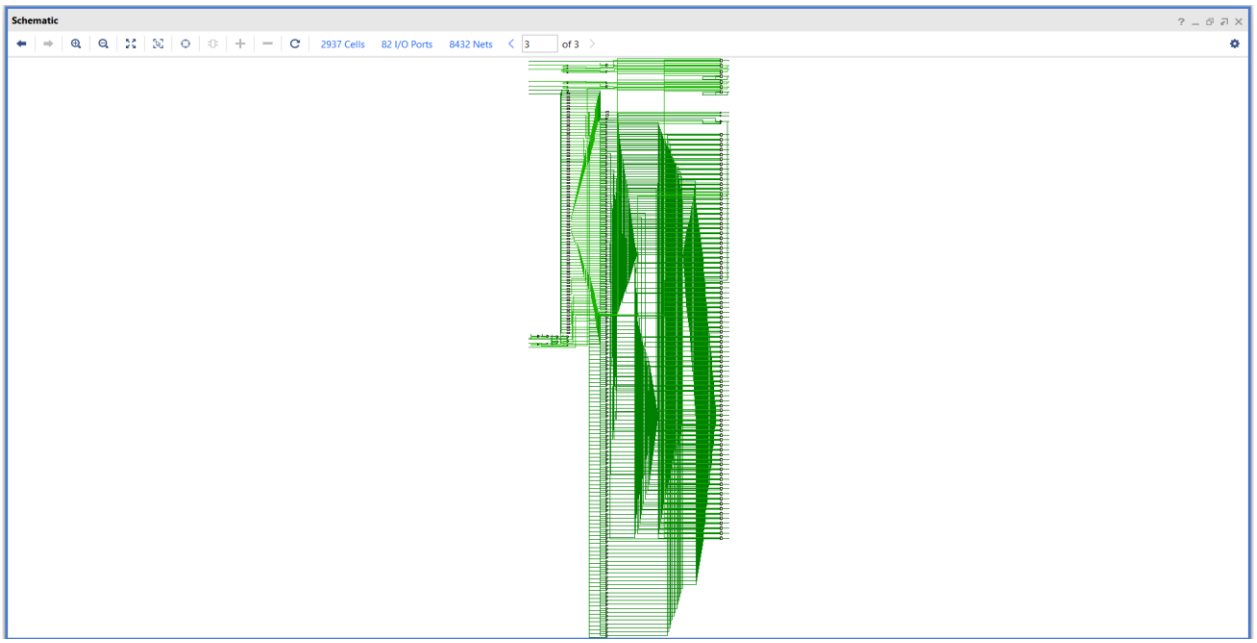
```
            end
            if (vif.ARVALID && vif.ARREADY) begin
                tr = new();
                tr.addr = vif.ARADDR;
                tr.is_write = 1'b0;
                mon_mbx.put(tr);
            end
        end
    endtask
endclass

class scoreboard;
    mailbox scb_mbx;
    transaction tr;

    task run();
        forever begin
            scb_mbx.get(tr);
            if (tr.is_write) begin
                $display("Write Transaction: Addr = %h, Data = %h", tr.addr,
tr.data);
            end else begin
                $display("Read Transaction: Addr = %h", tr.addr);
            end
        end
    endtask
endclass


module tb;
    logic ACLK;
```

```
logic ARESETn;

axi4_lite_if vif(ACLK, ARESETn);

axi4_lite_slave dut (
    .ACLK(ACLK),
    .ARESETn(ARESETn),
    .AWADDR(vif.AWADDR),
    .AWVALID(vif.AWVALID),
    .AWREADY(vif.AWREADY),
    .WDATA(vif.WDATA),
    .WSTRB(vif.WSTRB),
    .WVALID(vif.WVALID),
    .WREADY(vif.WREADY),
    .BRESP(vif.BRESP),
    .BVALID(vif.BVALID),
    .BREADY(vif.BREADY),
    .ARADDR(vif.ARADDR),
    .ARVALID(vif.ARVALID),
    .ARREADY(vif.ARREADY),
    .RDATA(vif.RDATA),
    .RRESP(vif.RRESP),
    .RVALID(vif.RVALID),
    .RREADY(vif.RREADY)
);

initial begin
    ACLK = 0;
    forever #5 ACLK = ~ACLK;
```

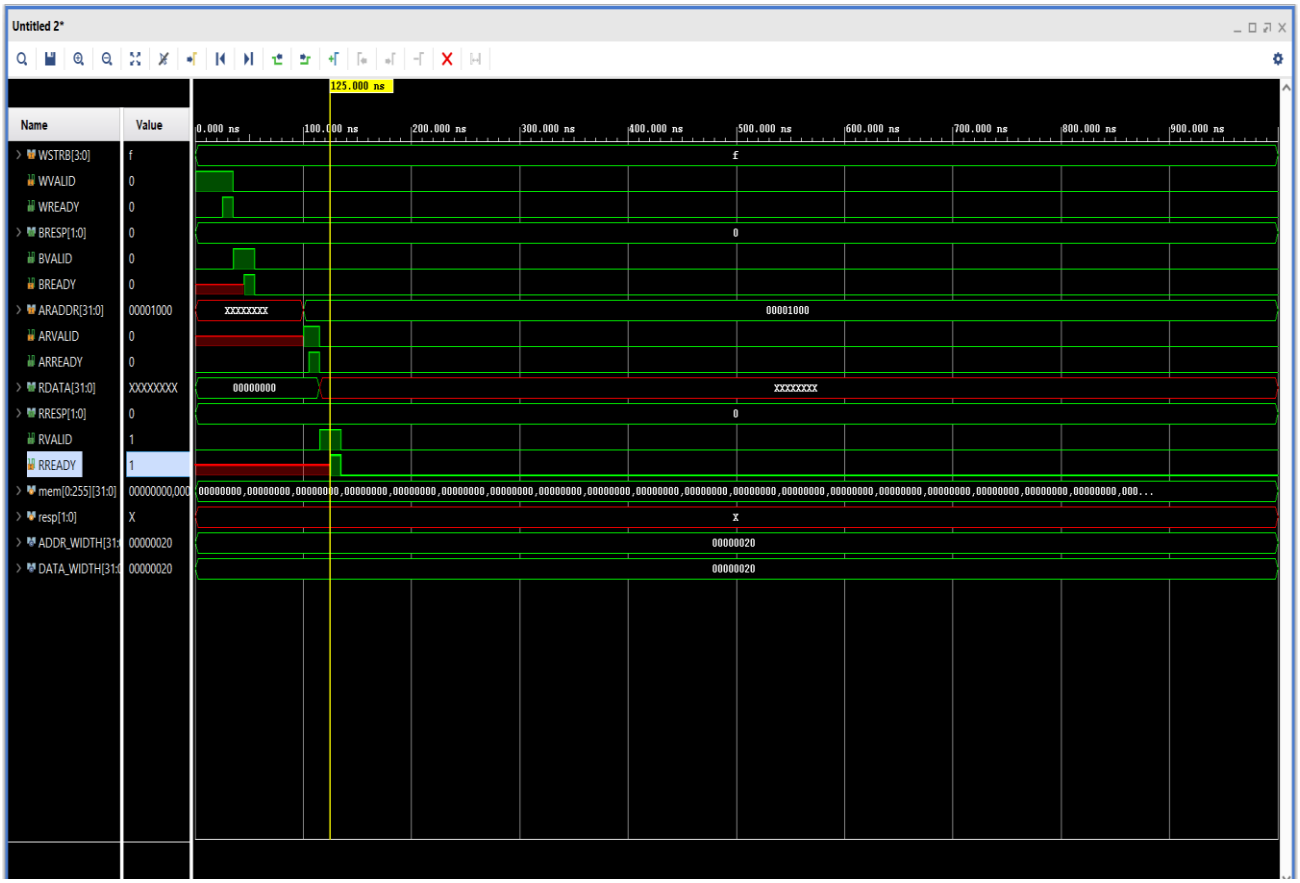
```
end

initial begin
    ARESETn = 0;
    #20;
    ARESETn = 1;
end

initial begin
    driver drv;
    monitor mon;
    scoreboard scb;
    mailbox drv_mbx, mon_mbx, scb_mbx;

    drv = new();
    mon = new();
    scb = new();

        drv_mbx = new();
    mon_mbx = new();
    scb_mbx = new();

    drv.drv_mbx = drv_mbx;
    mon.mon_mbx = mon_mbx;
    scb.scb_mbx = scb_mbx;


        drv.vif = vif;
    mon.vif = vif;
    // Generate transactions and send to driver
```

```
    fork
        begin
            transaction tr;
            tr = new();
            tr.addr = 32'h0000_1000; // Example address
            tr.data = 32'h1234_5678; // Example data
            tr.strb = 4'b1111;      // All bytes valid
            tr.is_write = 1'b1;     // Write transaction
            drv_mbx.put(tr);

            #100; // Wait for write to complete

            tr = new();
            tr.addr = 32'h0000_1000; // Example address
            tr.is_write = 1'b0;       // Read transaction
            drv_mbx.put(tr);
        end



            drv.run();
            mon.run();
            scb.run();
        join

    end
endmodule
```

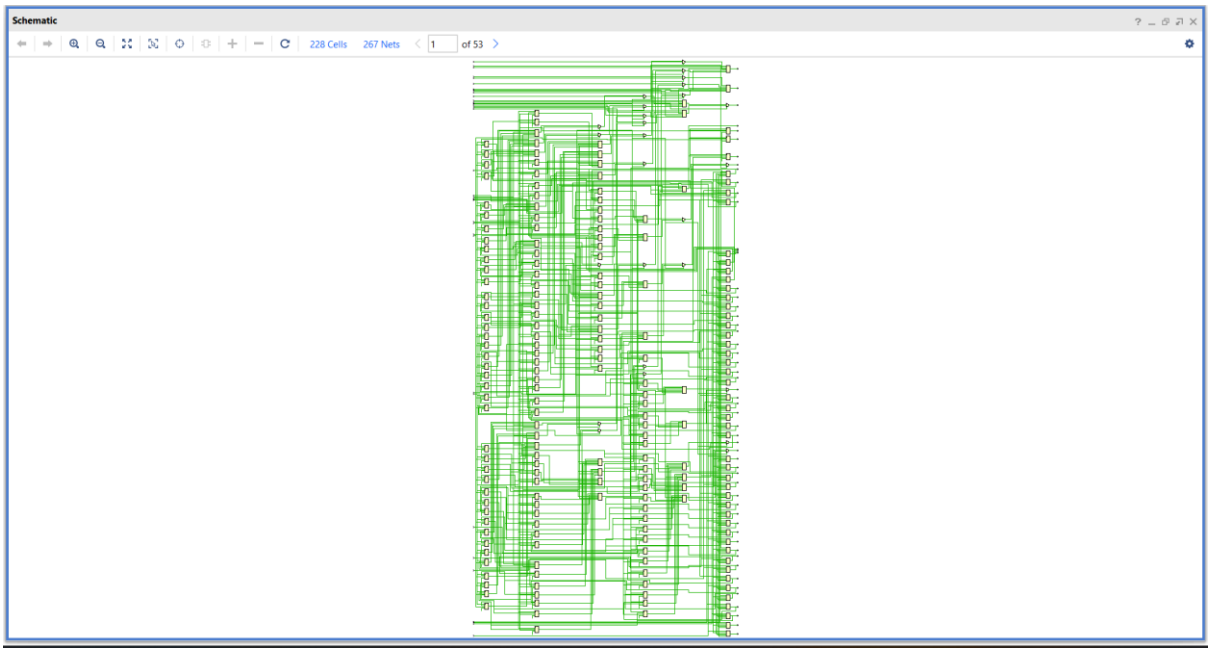
Elaborated Design:



Simulation:



Synthesis:



4. Advantages and Disadvantages

Advantages of AXI4-Lite:

- Simple Implementation – No burst support, reducing complexity.
- Efficient Resource Utilization – Uses fewer logic gates compared to full AXI4.
- Independent Read/Write Channels – Enables concurrent transactions.
- Handshaking Mechanism – Ensures data integrity with VALID and READY signals.
- Scalability – Easily integrated into AXI-based SoCs.

Disadvantages of AXI4-Lite:

- No Burst Transactions – Only supports single data transactions, limiting bandwidth.
- Higher Latency for Large Transfers – Since it doesn't support bursts, multiple transactions take longer.
- Limited Throughput – Designed for low-speed control applications rather than high-speed data transfer.

Despite these limitations, AXI4-Lite is widely used in embedded systems and peripheral control applications due to its ease of use and efficient design.

5. Applications of AXI4-Lite

The AXI4-Lite protocol is used in various embedded systems, FPGA designs, and ASIC implementations where simple register-based communication is required. Some common applications include:

1. Peripheral Interfaces – Used for GPIO, UART, SPI, I2C controllers in SoC designs.
2. Memory-Mapped Registers – Controls configuration registers in embedded devices.
3. Low-Speed Data Transfer – Suitable for slow, memory-mapped transactions in processors.
4. Embedded Processors – Implemented in ARM-based SoCs for control registers.
5. FPGA-Based System Designs – Utilized in Xilinx & Intel FPGAs for processor-to-peripheral communication.

AXI4-Lite is a critical component of modern SoC architectures, ensuring efficient register-based communication while maintaining compatibility with the AMBA AXI ecosystem.

6. Conclusion

This project successfully implements an AXI4-Lite Slave module using Verilog/SystemVerilog, along with testbench verification, simulation, and synthesis. The structured design approach ensures efficient memory-mapped communication between the master and the slave. Future enhancements can include:

- Extending the design to AXI4 with burst transactions
- Implementing an AXI4-Lite Master module
- Enhancing verification with UVM-based testbenches

---- The End----